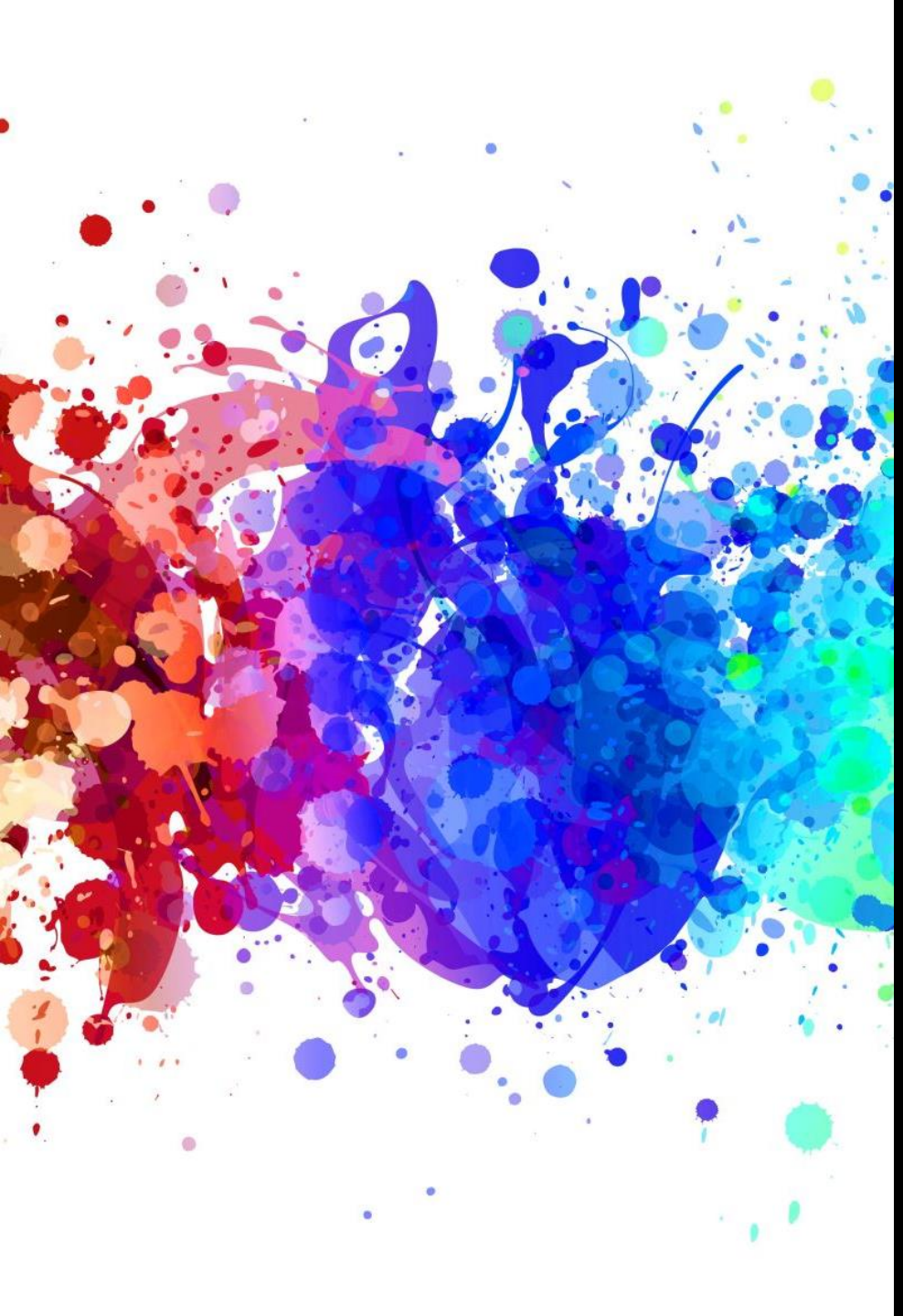


CAR DAMAGE DETECTION

Do you have a Car that is damaged? We will detect it!

GROUP MEMBERS

- Sakshi (C0908000)
- Bansil Patel (C0912873)
- Harsh Mohile (C0912872)
- Meet Patel (C0910378)
- Rachit Bhatt (C0902810)

CONTENTS

Introduction

Why Car Damage Detection?

Objective of the project

About Dataset

Training the Model

Results

Challenges

Improvement

Future work

Weekly Progress of the project

Conclusion

INTRODUCTION

- This project focuses on automating the detection and classification of car damage using deep learning techniques, specifically a VGG16 CNN.
- **Key Benefits**: Reduces human error, speeds up insurance claim processing, and helps repair shops in assessment.

WHY CAR DAMAGE DETECTION?

- **Current Issues**: Manual damage assessment is slow and can lead to inconsistent results.
- **Proposed Solution**: Using deep learning models like VGG16 can provide a consistent, fast, and scalable solution for car damage detection.

OBJECTIVE OF THE PROJECT

- The number of car user increases that eventual increase in the damage cars as well. So, our motive is to curtail the load in checking the fake damage parts or false claim and many more increase in the market.
- We try this to achieve in predicting the damage parts of the car using the VGG16 and we got the desired result as we want for the project with keeping the future reference in mind.

ABOUT DATASET

•Images: A collection of car images showing varying damage types (e.g., dents, scratches, cracks, etc.).
•Source: Public datasets (such as Car Damage Dataset), or custom datasets obtained from sources like insurance companies or repair shops.
•Labels: The dataset is annotated with labels indicating different damage categories (e.g., “minor", “moderate", “severity").

•**Preprocessing**:

- **Image Resizing**: All images are resized to 256*256 pixels to match VGG16’s input size.
- **Data Augmentation**: Techniques like rotation, flipping, zooming, and shifting are applied to generate more diverse training data and prevent overfitting.

TRAINING THE MODEL

Training Process:

- **Optimizer**: Adam (learning rate: 0.0001).
- **Loss Function**: Categorical Cross-Entropy.
- **Metrics**: Accuracy, Precision, Recall, F1-score.
- **Regularization**: Dropout & Batch Normalization.
- **Epochs & Batch Size**: 20-50 epochs, batch size of 32.

4 PHASES

Phase 1 – Car Detection:

- Does car exist?

Phase 2 – Car Damage Detection:

- Is car damaged?

Phase 3 – Car Damage Location:

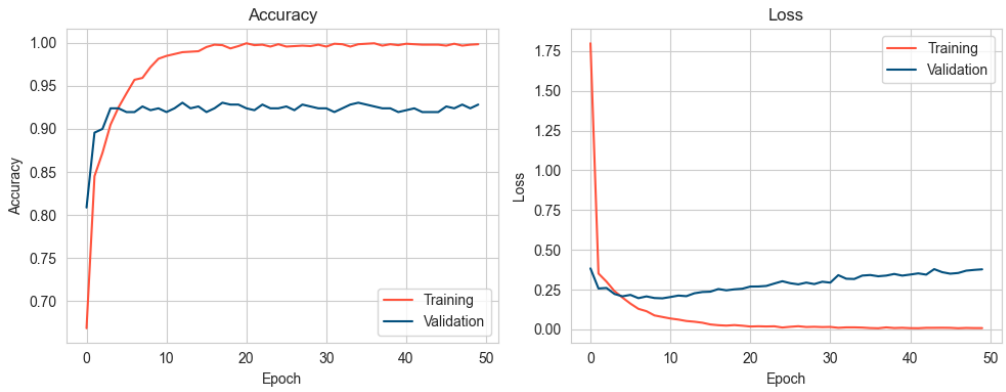
- Where is it damaged?

Phase 4 – Car Damage Severity:

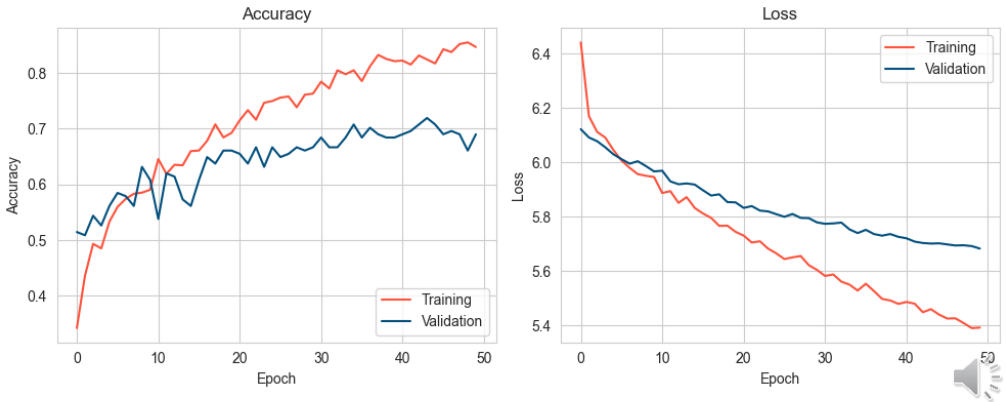
- How severe is it damaged?

PHASE 2 – DETECTING DAMAGE

FINETUNE MODEL



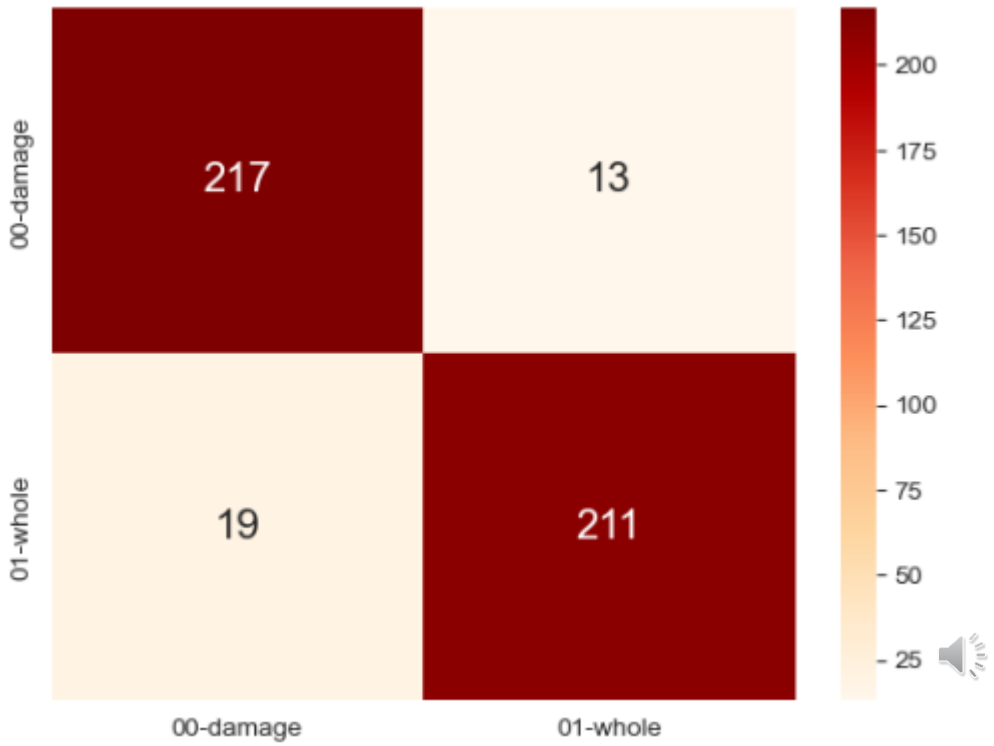
L2 REGULARISATION



PHASE 2 – IS CAR DAMAGED?

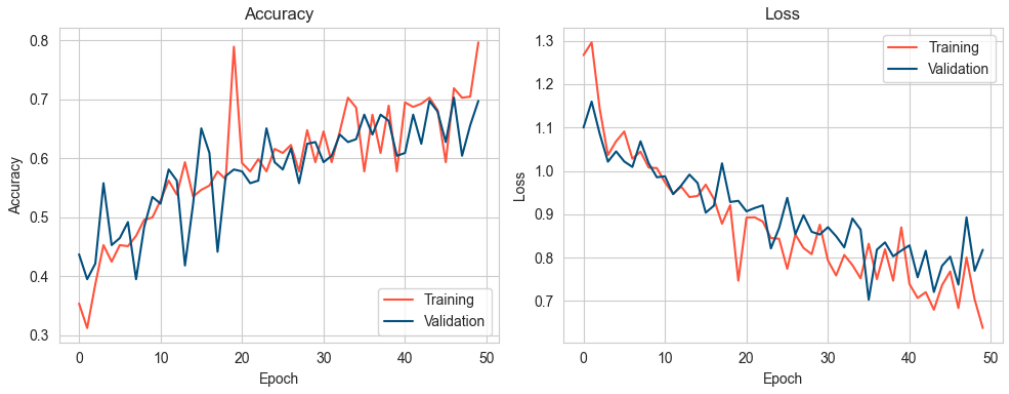
```
Found 460 images belonging to 2 classes.
8/8 ━━━━━━━━━━━━━━━━━━━━ 28s 3s/step

              precision    recall  f1-score   support

           0       0.93      0.93      0.93       230
           1       0.93      0.93      0.93       230

    accuracy                           0.93       460
   macro avg       0.93      0.93      0.93       460
weighted avg       0.93      0.93      0.93       460
```

Accuracy: 93%

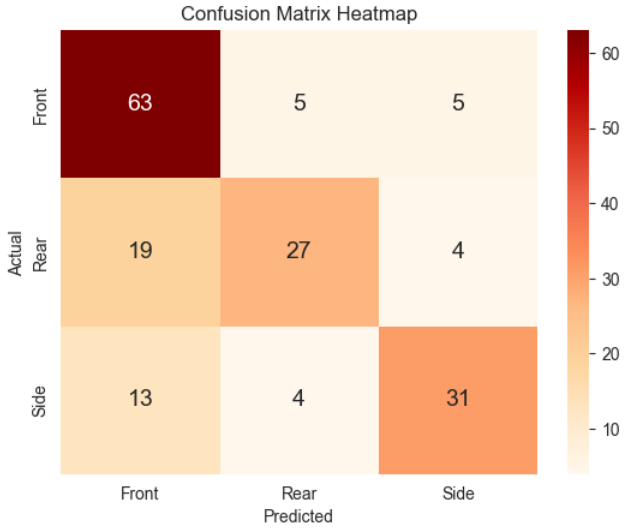


PHASE 3 – DAMAGE LOCATION

ACCURACY AND LOSS

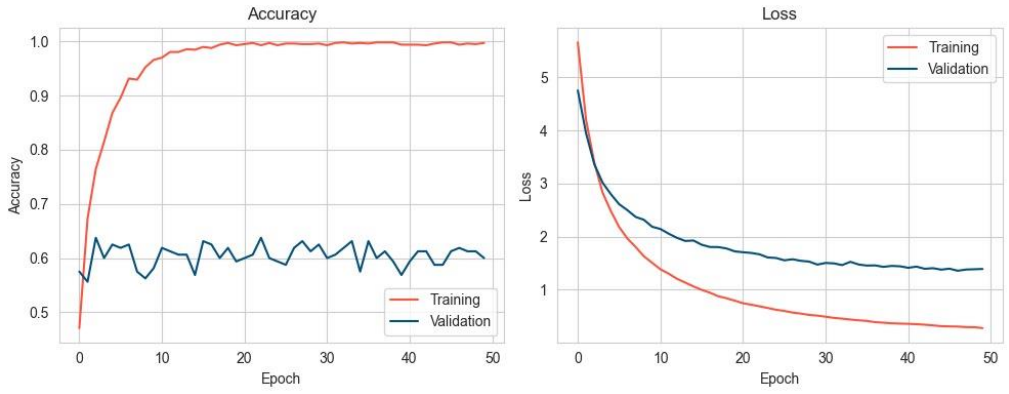


CONFUSION MATRIX

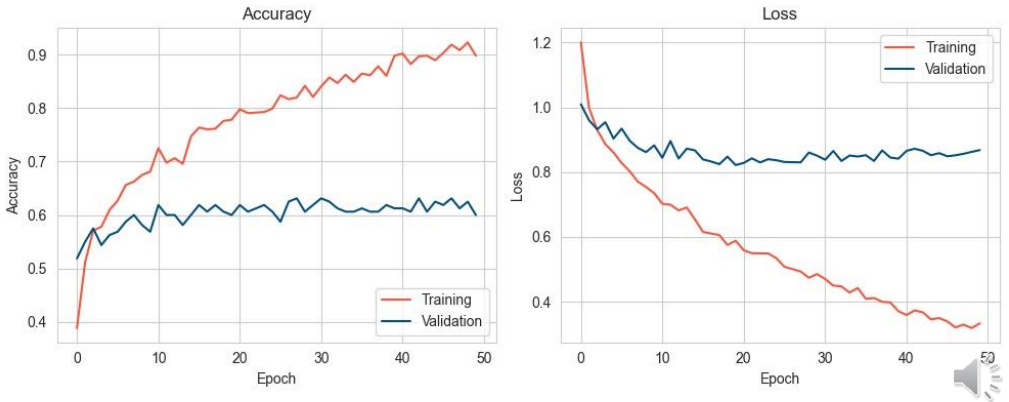


PHASE 4 – DAMAGE SEVERITY

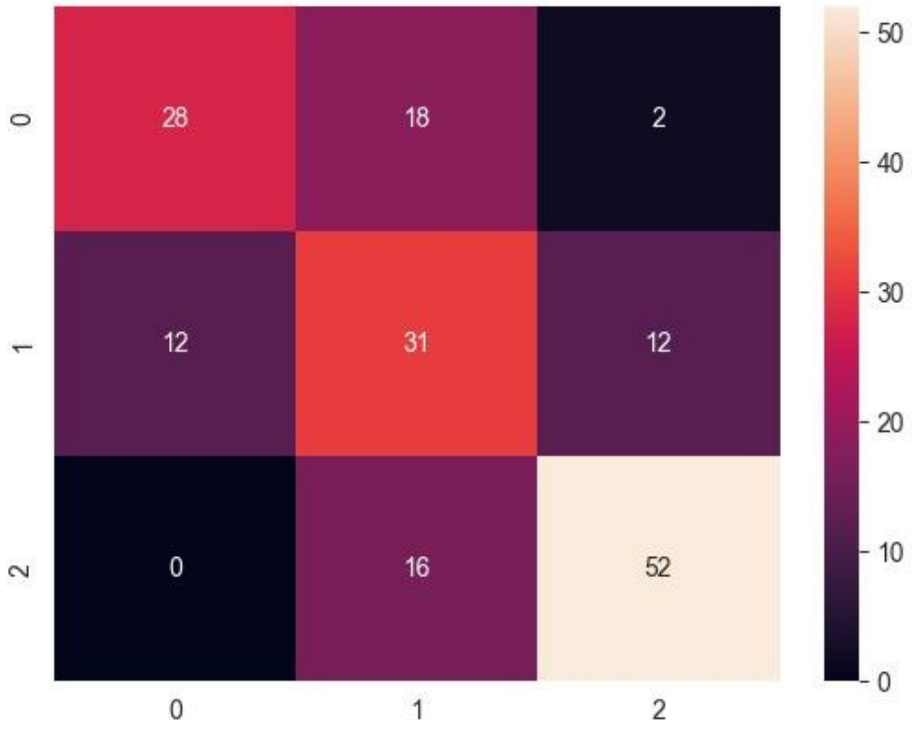
ACCURACY AND LOSS



TUNED MODEL



PHASE 4 CONT. – METRICS



	precision	recall	f1-score	support
0	0.70	0.58	0.64	48
1	0.48	0.56	0.52	55
2	0.79	0.76	0.78	68
accuracy			0.65	171
macro avg	0.65	0.64	0.64	171
weighted avg	0.66	0.65	0.65	171

Accuracy: 65%

DEPLOYMENT

USING **FLASK** FRAMEWORK – LIGHTWEIGHT.

USED 4 MODELS USING `.KERAS` FOR EACH PHASE.

RESULTS

In this 2 images we can see that one images is of whole car and other image is of damaged car .

1. For whole car image we can see that the model is giving false for every feature
2. In second image we can see that car is damage and our model is predicting accurately like the location is rear, damage severity is severe

More test result are been shown in the next slide

Damage Assessment Complete

Results:

Car validation check	
Damage validation check	✖
Location	Unknown
Severity	Unknown

Damage Assessment Complete

Results:

Car validation check	✔
Damage validation check	
Location	Rear
Severity	Severe

IN THE LEFT SIDE OF IMAGE THE SEVERITY IS MINOR
IN THIS SLIDE OF IMAGE THE MIDDLE IMAGE THE SEVERITY IS MINOR
IN THE RIGHT SIDE OF IMAGE THE SEVERITY IS SEVERE

Damage Assessment Complete

Results:

Car validation check	✔
Damage validation check	✔
Location	Rear
Severity	Minor

Damage Assessment Complete

Results:

Car validation check	✔
Damage validation check	✔
Location	Side
Severity	Minor

Damage Assessment Complete

Results:

Car validation check	✔
Damage validation check	✔
Location	Front
Severity	Severe

CHALLENGES

•**Variety of Damages**: The model needs to detect both subtle and major damage, which can look very different.
•**Image Quality**: Low-quality images or poor lighting conditions can affect the model's ability to detect damage accurately.
•**Class Imbalance**: If one type of damage is more frequent in the dataset, the model might become biased toward that class.

IMPROVEMENT

- **Data Augmentation**: Implementing more complex augmentation techniques like color shifting, cropping, or synthetic data generation.
- **Ensemble Methods**: Combine VGG16 with other models (like ResNet or MobileNet) to increase robustness and reduce errors.

WEEKLY PROGRESS

Weekly Timeframe	Descripition	Leader
Week-1	Finding appropriate project – Fetching the Dependencies/Modules.	Rachit
Week-2	Fetching Dataset and Configuring GitHub.	Sakshi
Week-3	Image Analysis and Agile.	Harsh
Week-4	Utilizing VS Code for Development.	Meet
Week-5	Exploring Dataset.	Bansil
Week-6	Accomplished Phase 1.	Meet
Week-7	Accomplished Phase 2.	Sakshi
Week-8	Accomplished Phase 3.	Rachit
Week-9	Accomplished Phase 4.	Harsh
Week-10	Deployment using Flask.	Bansil
Week-11	Integrating Models to Deployed Project.	Meet

FUTURE WORK

REPAIR COST ESTIMATION

DEALING WITH MORE CAR COMPONENTS

CONCLUSION

- The project demonstrates how deep learning, particularly VGG16, can be applied to the task of car damage detection from images.
- The model successfully identifies different types of damage, making it useful for industries like insurance and vehicle repair.